

Autonomous Emergency Response Robot

IEEE SoutheastCon Hardware Competition

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SoutheastCon Hardware Competition



- Theme:
 - Natural disasters can devastate whole countries.
 - Fast emergency response is critical for preventing further crisis and saving lives.

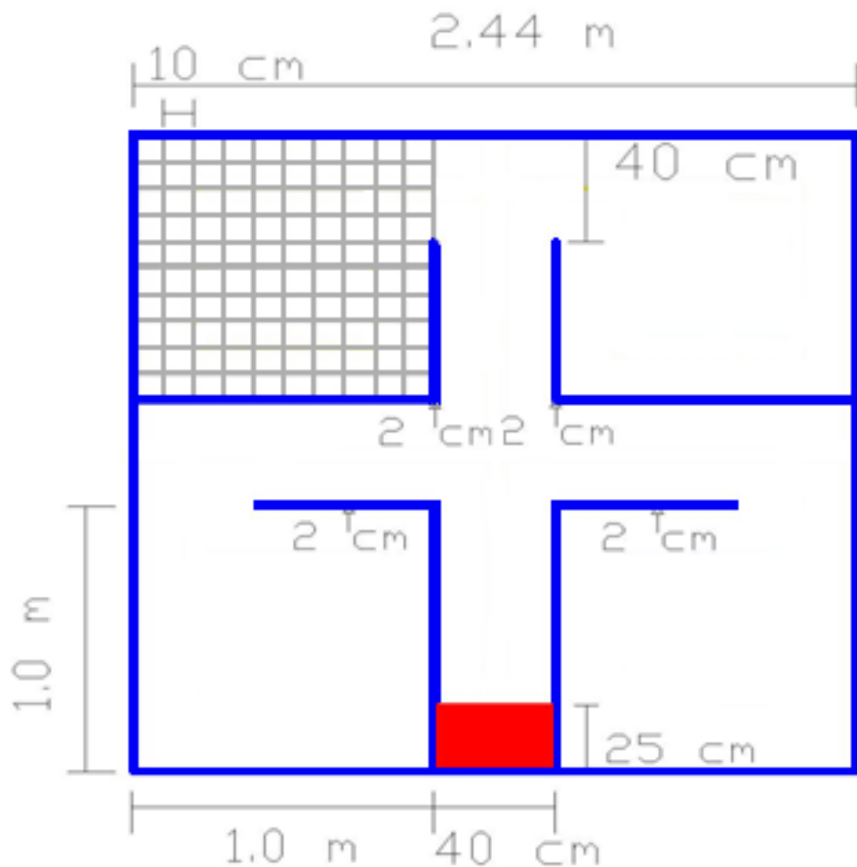
Design Specifications

- Purpose:
 - Build an autonomous emergency response robot that can safely evaluate the situation and relay vital information to emergency response teams.
- Goals:
 - Navigate through specified floor plan
 - Identify and avoid obstacles
 - Sense hazards
 - Find and determine status of victims
 - Report vital information to emergency responders
- Restrictions:
 - Size: 20 x 20 x 20 cm
 - Time Limit: 4 minutes
 - Robot can not jump, fly, hover, or climb over walls.
 - Wireless communication is not allowed.

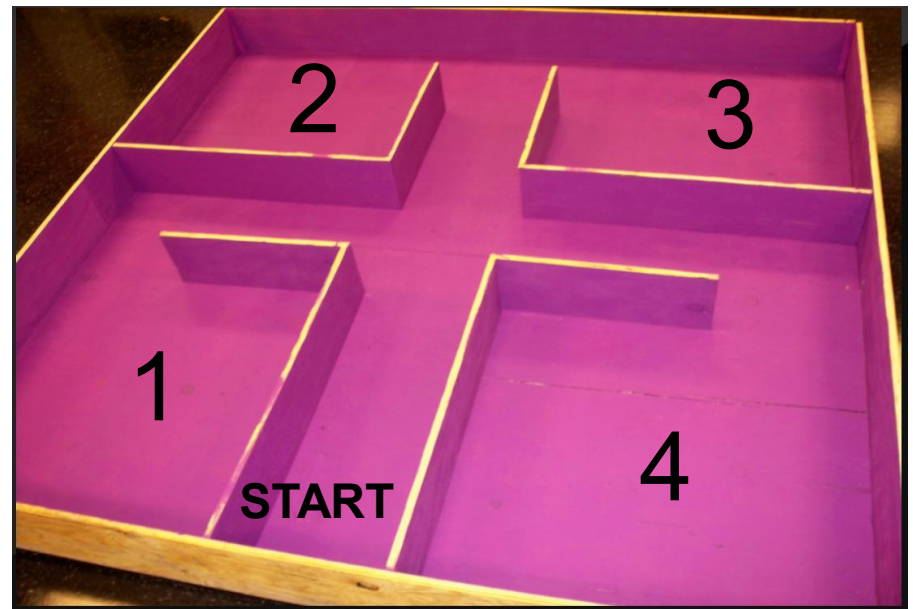
Scoring

Category	Points	Total Possible
Audio Output	150	150
Digital Display	50	50
Victim Position*	60	720
Victim Status	40	480
Hazard Location	200	200
Total		1,600

Course Layout



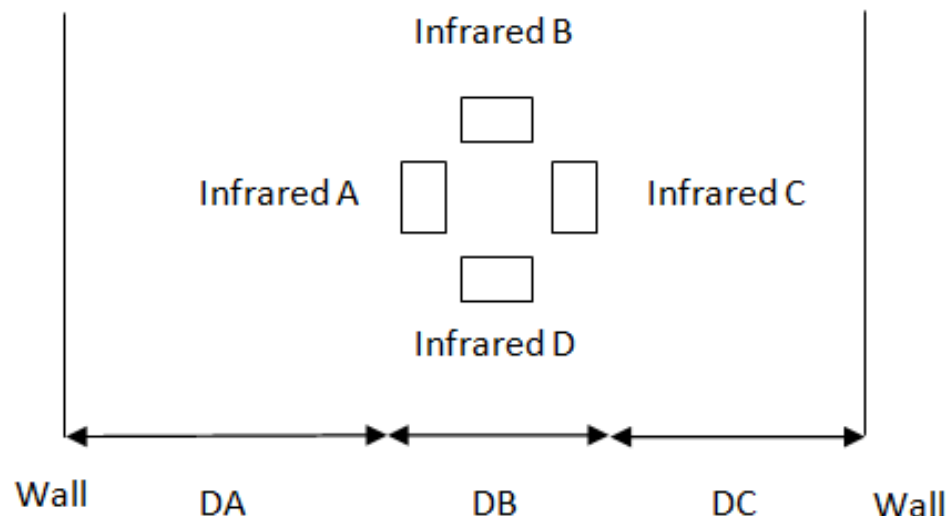
- RED: designated starting area
- 10 x 10 grid for victims and hazards
- obstacles can be placed at any angle in hallways and rooms



Navigation Platform

- Platform has the same orientation even when body rotates.
 - B = North
 - A = West
 - C = East
 - D = South
- Distance measurements used in calibration:
 - DB = constant 2 cm
 - $DA + DB + DC = \text{hallway width or distance between room walls}$

Figure 2: Top View of Robot Showing Infrared Sensors Mounted on Servo

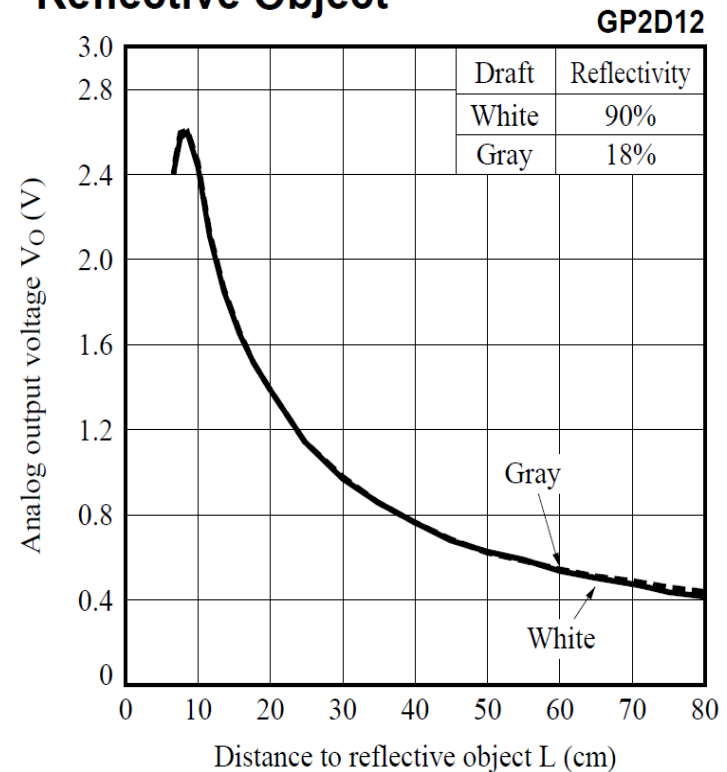


Calibrating Sensors

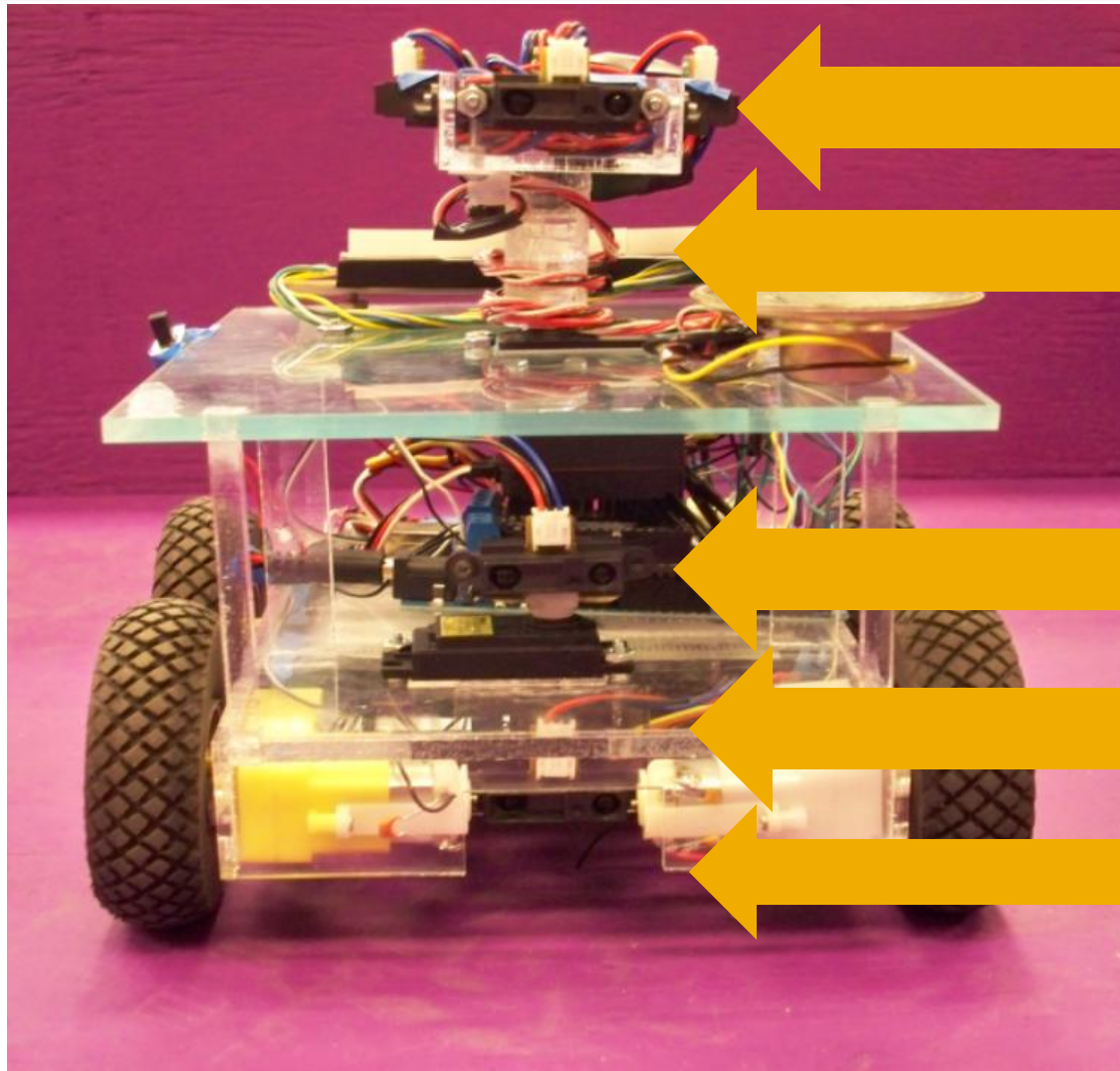
- The IR sensors sit further inward to prevent the spike in error between 0 cm and 8 cm.
- Voltage rating of 5V divided by 1024 points to give a value per step
 - $5V/1024 = 0.0048828125$
- Exponential
 - Slope: $1/\text{Volts} * 28$
 - Distance = $(\text{Volts} \times 28)$

- Arduino code:
 - `float volts = analogRead(IRpin)*0.0048828125;`
 - `float distance = 28*pow(volts, -1.0);`

Fig.6 Analog Output Voltage vs.Distance to Reflective Object



Obstacle Detection



Navigation Platform

360° Servo Motor

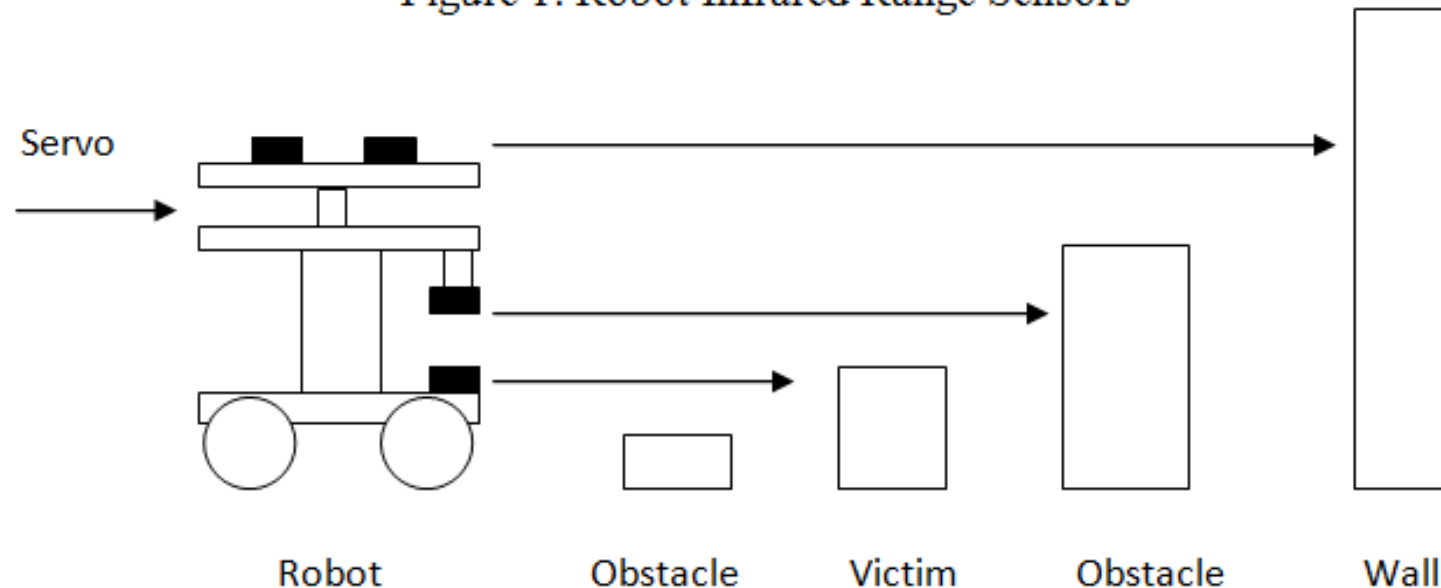
Obstacle Detector

180° Servo Motor

Victim Detector

Obstacle Avoidance

Figure 1: Robot Infrared Range Sensors



- Objects to Avoid:
 - Walls
 - 4 cm x 9 cm obstacles
 - 3 cm x 2 cm victims
 - 2 cm x 4 cm obstacles

Obstacle Avoidance

- The robot must know where the obstacles and the walls are to navigate through the course.
- The servo motor pans from 30° to 150° so the infrared sensor can take measurements.
- Values returned are distance measurements
(ex. [121, 83, 2, 2, 2, 2, 54, 85, 141])
where the value indicates how close the object is to the robot.
- Program analyzes the values returned and the angle of the servo motor to decide whether the robot can fit around the obstacle or if it needs to try a different route.



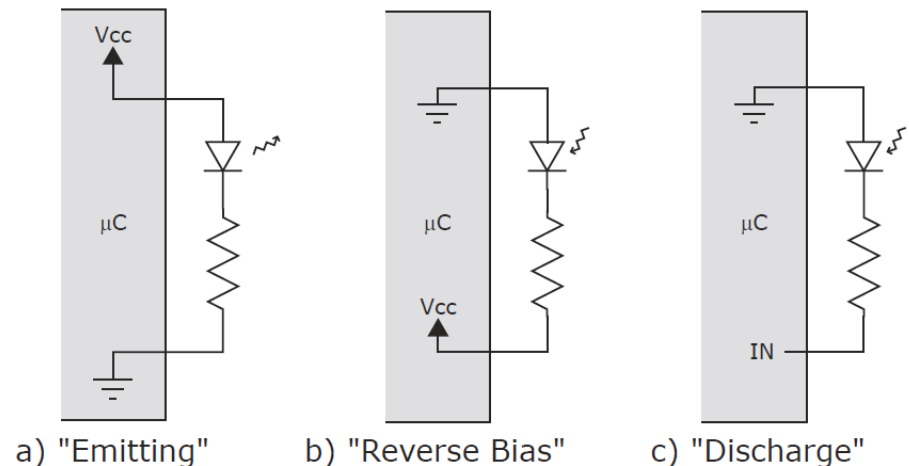
The image shows how the infrared sensor input to the microcontroller when mounted on a servo motor matches the actual placement of the obstacles.

Detecting Victims

- Victims: Green LED in PVC cap
 - ON = Conscious
 - OFF = Dead
 - No battery connections
 - Blinking = Unconscious
 - 2 Hz with 50% duty cycle
 - Blinks twice every second



- Detection: Green LED used to detect the victim green LED
 - a) Connect to two bidirectional I/O pins and drive LED with current in forward direction
 - b) Drive LED in the reverse-bias direction
 - c) Current discharges and is measured
- Time to discharge is inversely proportional to amount of incident light measured



Robot Output

Speaker

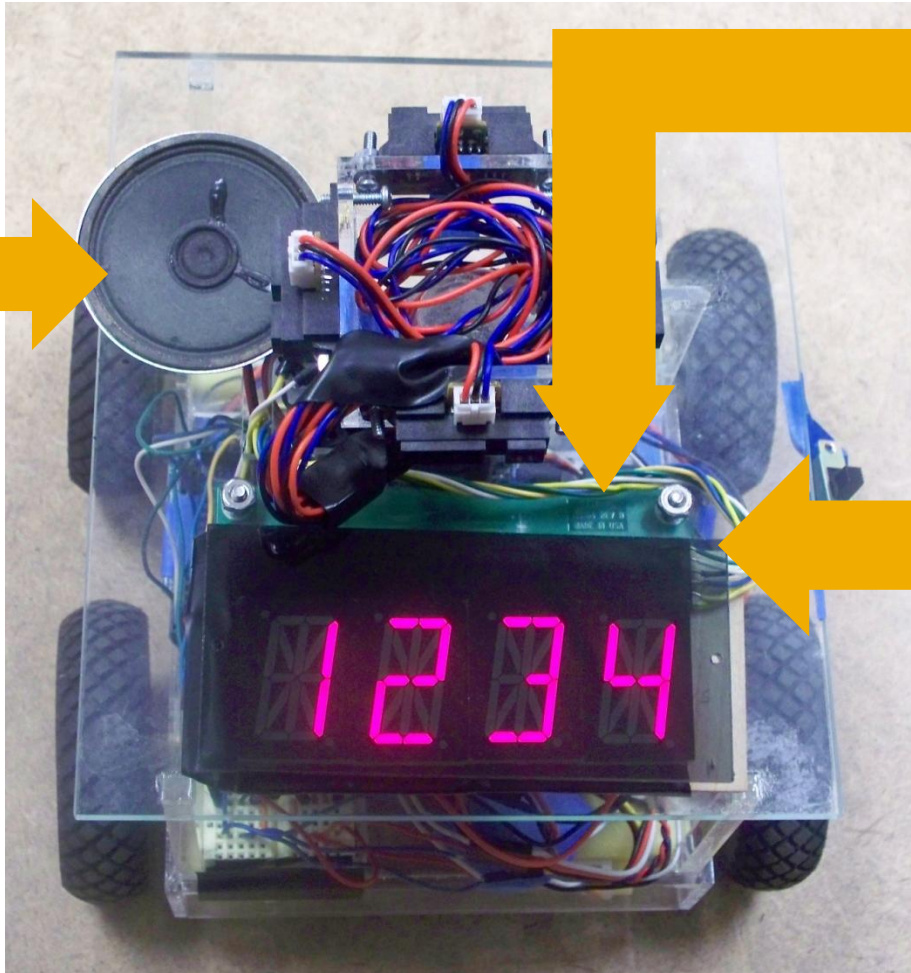
8 ohm

Audio Controller

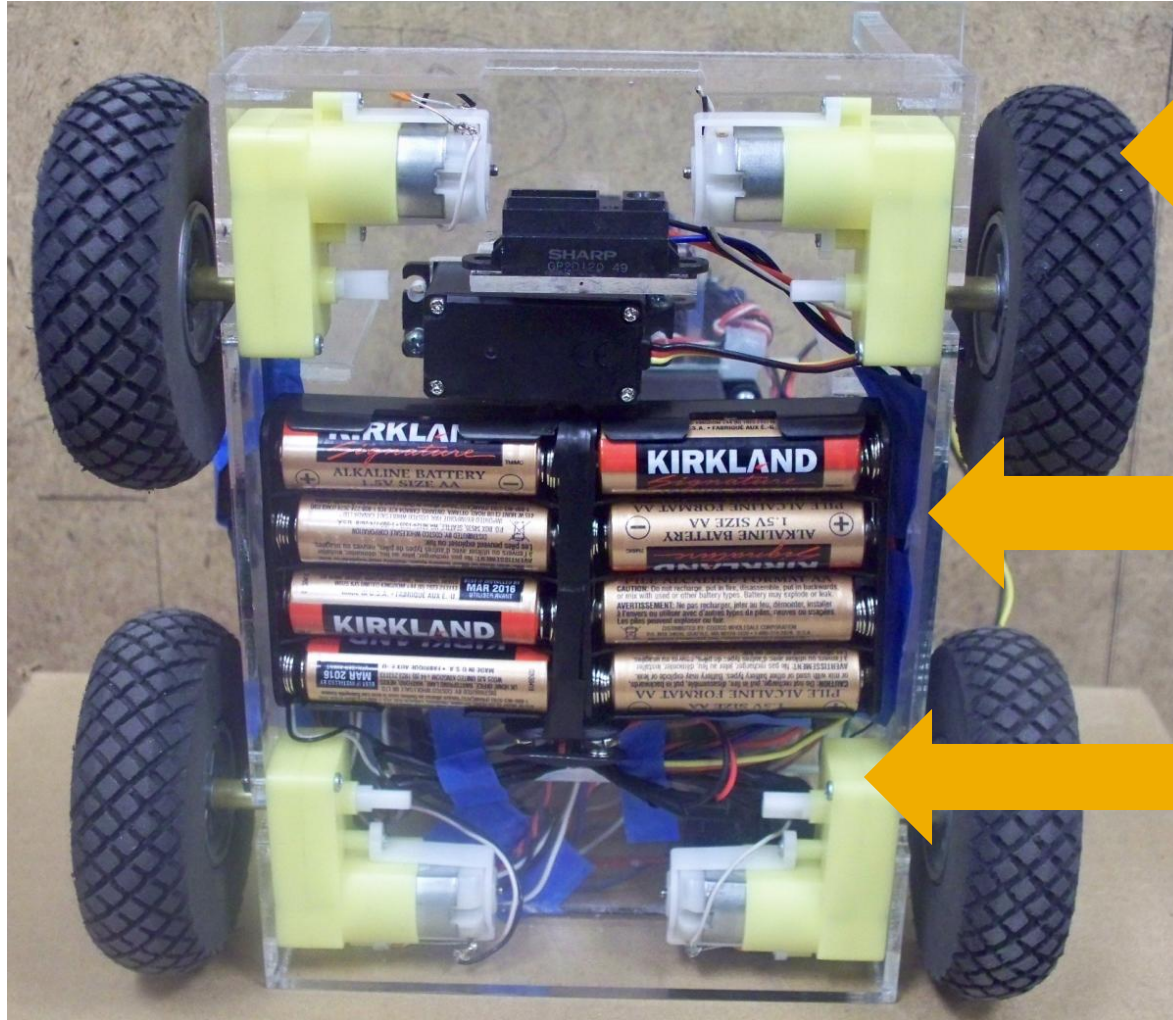
Voice synthesizer
Developed by RC Systems
Model: V8600A

Display

4 digit LED display
1" Alphanumeric numbers
Controlled by stackable serial drivers
Bright segments
Covered with antiglare material to
increase readability



Robot Body



Wheels

8 cm. diameter diamond-tread wheels

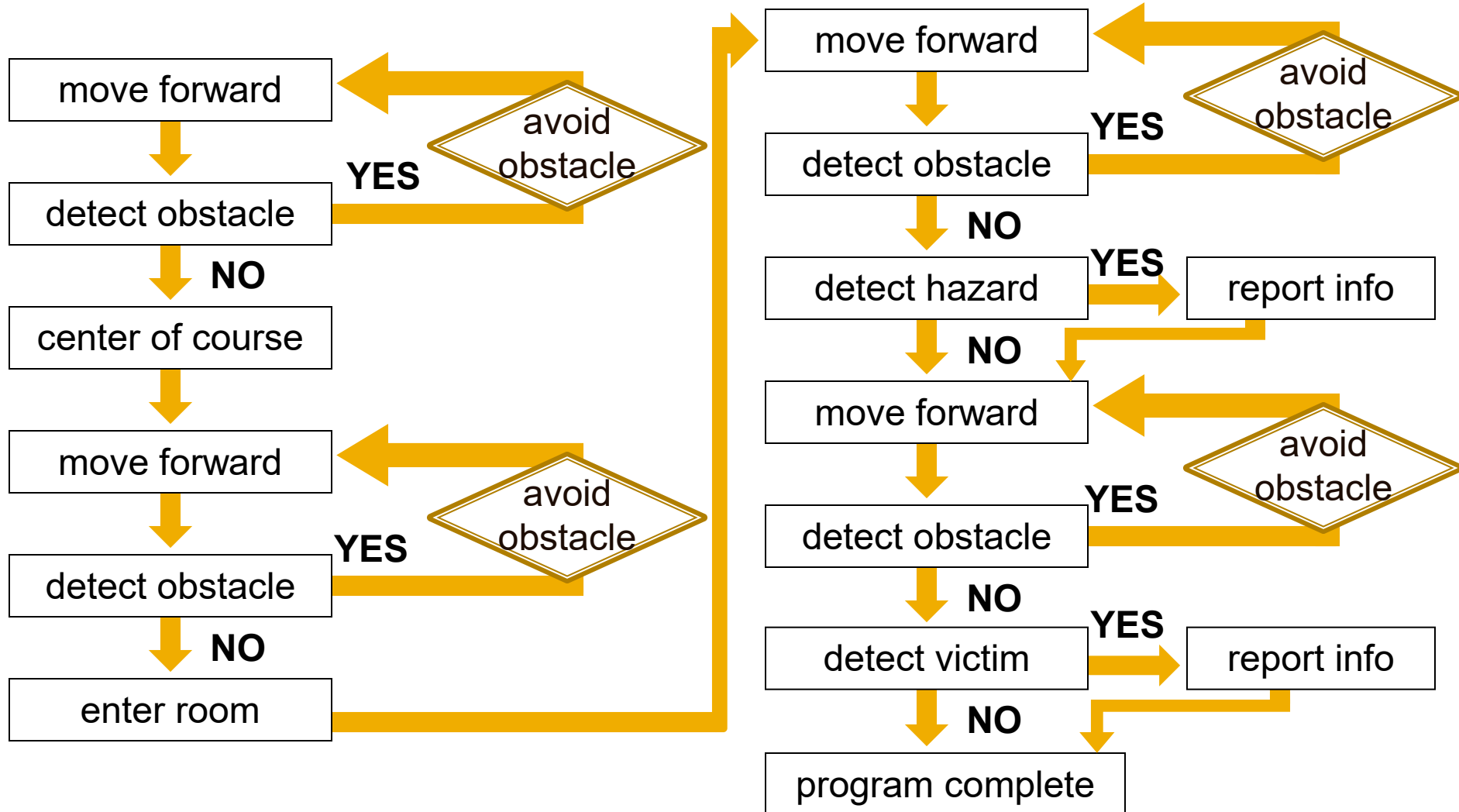
Battery Pack

8 AA batteries
Nominal voltage = 12V

Micro DC Geared Motors

Gear ratio: 1:120
No-load speed (6V): 180RPM
Stall current(6V):470mA
Torque (6V): 1.92Kgcm

Overall Program Design



Cost

Part	Cost
Wiring	\$10.00
SHARP Infrared Sensors (6)	\$24.00
Electromagnetic Field Detector	\$10.00
Green LEDs	\$2.00
V8600A Audio Synthesizer	\$100.00
LED Digital Display	\$30.00
Batteries	\$35.00
360 full rotation servo motor	\$13.50
180 servo motor	\$15.60
Robot Body and Motors	\$49.20
Arduino Mega Microcontroller	\$55.00
DFRobot 2A Motor Controller	\$17.00
4 x 8 sheets of plywood	\$120.00
1 gallon of purple matte paint	\$30.00
TOTAL	\$511.30

- Cost to reproduce
- Minus course materials
 - prototype cost = \$361.30
- Relatively low
 - Salvaged components from previous projects